3 Second Order Linear Ordinary Differential Equations: Initial Value Problems

Reference: Boyce & DiPrima, Chapter 3

Definition 3.1 A **second order linear ordinary differential equation** has the form

$$a_2(t)y'' + a_1(t)y' + a_0(t)y = G(t)\,, \tag{3.1}$$

where we have used a prime symbol $'$ to denote a derivative. The coefficients on the left-hand side (namely $a_2(t)$, $a_1(t)$ and $a_0(t)$), and the function $G(t)$ on the right-hand side, are specified (known) functions of the independent variable t. This is known as the **standard form** of a linear second order ODE. Alternatively, very often we deal with the equation in the **normalised form**

$$y'' + p(t)y' + q(t)y = g(t)\,. \tag{3.2}$$

Definition 3.2 The second order linear ordinary differential equation (3.2) is said to be **homogeneous** if the right-hand side $g(t)$ is zero for all t. Otherwise the equation is called **nonhomogeneous**. The term $g(t)$ is sometimes referred to as the nonhomogeneous term.

3.1 Fundamental set of solutions of linear homogeneous equations

In this section we set $g(t)$ to zero and we briefly describe some general theory for the second order linear homogeneous ODE

$$y'' + p(t)y' + q(t)y = 0\,. \tag{3.3}$$

When we have understood how to solve this homogeneous equation, we will be able to make progress in solving a nonhomogeneous equation.

Theorem 3.3 — Principle of superposition for homogeneous equations. If $y_1(t)$ and $y_2(t)$ are two solutions of the homogeneous equation (3.3) then

$$y_3(t) = c_1y_1(t) + c_2y_2(t)$$

is a solution of (3.3) for any constants c_1 and c_2.

Proof.

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Definition 3.4 The Wronski determinant, or **Wronskian** for short, of two functions $y_1(t)$ and $y_2(t)$ is the function

$$\begin{aligned} W(t) &= \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} \\ &= y_1(t)y_2'(t) - y_2(t)y_1'(t) \\ &= W[y_1, y_2](t) \end{aligned} \tag{3.4}$$

Theorem 3.5 Let y_1 and y_2 be solutions of

$$y'' + p(t)y' + q(t)y = 0\,,$$

where p and q are continuous on an open interval $I : \alpha < t < \beta$. Then y_1 and y_2 are **linearly independent** on I if and only if $W[y_1, y_2](t)$ is never zero in I.

Theorem 3.6 — Abel's Identity. If y_1 and y_2 are solutions of

$$y'' + p(t)y' + q(t)y = 0\,,$$

where p and q are continuous on an open interval $I : \alpha < t < \beta$, then the Wronskian is given by

$$W[y_1, y_2](t) = c \exp\left[-\int p(t)\,dt\right], \tag{3.5}$$

where c is a certain constant that depends on y_1 and y_2, but not on t. Consequently, $W[y_1, y_2](t)$ is either zero for all t in I (if $c = 0$) or it is never zero in I (if $c \neq 0$).

Proof.

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Theorem 3.7 If y_1 and y_2 are two linearly independent solutions of

$$y'' + p(t)y' + q(t)y = 0\,, \tag{3.6}$$

where p and q are continuous on an open interval $I : \alpha < t < \beta$, then the family of solutions

$$y = c_1 y_1(t) + c_2 y_2(t) \tag{3.7}$$

with arbitrary constant coefficients c_1 and c_2 includes every solution of (3.6) and we call (3.7) the **general solution** of (3.6). The solutions y_1 and y_2 are said to form a **fundamental set of solutions** of (3.6) on I.

Example 3.8 Verify that the functions $y_1(t) = \cos(3t)$ and $y_2(t) = \sin(3t)$ satisfy the ODE

$$y'' + 9y = 0\,.$$

Do they constitute a fundamental set of solutions? Justify your answer. Also verify Abel's identity.

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3.2 Initial value problems

An **initial value problem** (IVP) for a second order linear differential equation comprises the differential equation and a pair of initial conditions

$$y(t_0) = y_0, \quad y'(t_0) = y_0', \tag{3.8}$$

where y_0 and y_0' are given numbers. We note that the initial conditions prescribe not only the particular point (t_0, y_0) through which the graph of the solution must pass, but also the slope y_0' of the graph at that point.

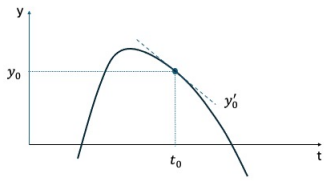


Figure 3.1: The initial condition.

Theorem 3.9 Consider the initial value problem

$$y'' + p(t)y' + q(t)y = g(t), \quad y(t_0) = y_0, \quad y'(t_0) = y_0', \tag{3.9}$$

where p, q and g are continuous on the open interval $I : \alpha < t < \beta$, with $t_0 \in I$. Then there exists a unique solution $y = \phi(t)$ of the IVP throughout the interval I.

Example 3.10 Without solving the IVP

$$(t^2 - 3t)y'' + t\,y' - (t+3)\,y = 0, \quad y(1) = 2, \quad y'(1) = 1\,,$$

find the longest interval in which the solution is guaranteed to exist.

Example 3.11 Without solving the IVP

$$(t-2)y'' + t\,y' + (\ln|t|)\,y = 0, \quad y(1) = 0, \quad y'(1) = 1\,,$$

find the longest interval in which there is certain to be a unique solution.

3.3 Homogeneous equations with constant coefficients[5]

[5] These ODEs have been seen before in the MTH1002, part 2, lecture notes.

Consider the second order ODE (3.1) in the homogeneous case (in which $G(t) \equiv 0$) and when the coefficients a_2, a_1 and a_0 are constants. In this case (3.1) can be written as

$$ay'' + by' + cy = 0\,, \tag{3.10}$$

with $a \neq 0$.

We postulate a solution of the form

$$y = e^{mt}\,, \tag{3.11}$$

where m is a constant that is to be determined. We calculate the derivatives

$$y' = me^{mt}, \quad y'' = m^2 e^{mt}$$

and substitute into (3.10) to obtain the **characteristic equation**

$$am^2 + bm + c = 0\,. \tag{3.12}$$

There are two roots of this characteristic equation:

$$m_{1,2} = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}\,.$$

They are:

- Real and different when $b^2 - 4ac > 0$, giving $m_1 \neq m_2$.
- Complex conjugates when $b^2 - 4ac < 0$, giving $m_1 = \lambda + i\mu$ and $m_2 = \lambda - i\mu$.
- Real and repeated when $b^2 - 4ac = 0$, giving $m_1 = m_2$.

3.3.1 Real and different roots

The two solutions of (3.10) are

$$y_1(t) = e^{m_1 t}, \quad y_2(t) = e^{m_2 t}\,, \quad m_1 \neq m_2\,.$$

The Wronskian

$$W[y_1, y_2](t) = e^{m_1 t} m_2 e^{m_2 t} - e^{m_2 t} m_1 e^{m_1 t} = (m_2 - m_1) e^{(m_1 + m_2)t} \neq 0 \quad \forall t$$

since the roots are different, i.e. $m_1 \neq m_2$

Thus the solutions are linearly independent for all t and by Theorem 3.7 the **general solution** is

$$y = c_1 e^{m_1 t} + c_2 e^{m_2 t}, \quad t \in \mathbb{R}\,. \tag{3.13}$$

Activity 3.12 Verify that the linear sum (3.13) is a solution of (3.10) by differentiating and substituting.

Activity 3.13 Now suppose that we want to find a particular member of the family of solutions (3.13) that satisfies the initial conditions

$$y(t_0) = y_0, \quad y'(t_0) = y_0'.$$

Find c_1 and c_2.

Answer: $c_1 = \frac{y_0 m_2 - y_0'}{m_2 - m_1} \exp(-m_1 t_0)$, $c_2 = \frac{y_0' - y_0 m_1}{m_2 - m_1} \exp(-m_2 t_0)$

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Example 3.14 Find the general solution of

$$y'' + 5y' + 6y = 0\,.$$

Example 3.15 Find the solution of the initial value problem

$$y'' + 5y' + 6y = 0\,, \quad y(0) = 2, \quad y'(0) = 3.$$

Graph the solution and describe its behaviour as t increases.

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In summary, when the roots are real and different ($m_1 \neq m_2$) the general solution is a sum of exponentials. As t increases, the magnitude of the solution either tends to zero (when both exponents are negative), grows rapidly (when at least one exponent is positive), or approaches a constant (when one exponent is zero and the other is negative).

3.3.2 Complex conjugate roots

The two solutions of (3.10) are

$$y_1(t) = e^{(\lambda+i\mu)t}, \quad y_2(t) = e^{(\lambda-i\mu)t},$$

where λ and μ are real with $\mu \neq 0$.

Recalling Euler's identity

$$e^{it} = \cos t + i \sin t \tag{3.14}$$

we have

$$y_1(t) = e^{\lambda t}e^{i\mu t} = e^{\lambda t}[\cos(\mu t) + i\sin(\mu t)]$$

and

$$y_2(t) = e^{\lambda t}e^{-i\mu t} = e^{\lambda t}[\cos(-\mu t) + i\sin(-\mu t)] = e^{\lambda t}[\cos(\mu t) - i\sin(\mu t)].$$

Hence we conclude that $y_1(t)$ and $y_2(t)$ are complex-valued solutions of (3.10).

Now note that, for example,

$$y_1(t) = e^{\lambda t}\cos(\mu t) + ie^{\lambda t}\sin(\mu t) = u + iv, \tag{3.15}$$

say, where u and v are real. Substituting into (3.10) we obtain

$$[au'' + bu' + cu] + i[av'' + bv' + cv] = 0.$$

However, when a complex number is zero both its real part and its imaginary part must be zero. Hence

$$au'' + bu' + cu = 0 \quad \text{and} \quad av'' + bv' + cv = 0,$$

that is, both $u = e^{\lambda t}\cos(\mu t)$ and $v = e^{\lambda t}\sin(\mu t)$ are real valued solutions of (3.10).

The Wronskian (see (3.4))

$$\begin{aligned} W[u,v](t) &= uv' - vu' \\ &= e^{\lambda t}\cos(\mu t)[e^{\lambda t}\sin(\mu t)]' - e^{\lambda t}\sin(\mu t)[e^{\lambda t}\cos(\mu t)]' \\ &= e^{\lambda t}\cos(\mu t)[e^{\lambda t}\mu\cos(\mu t) + \lambda e^{\lambda t}\sin(\mu t)] \\ &\quad -e^{\lambda t}\sin(\mu t)[-\mu e^{\lambda t}\sin(\mu t) + \lambda e^{\lambda t}\cos(\mu t)] \\ &= e^{2\lambda t}\mu, \end{aligned} \tag{3.16}$$

which is non-zero for all t and hence u and v are linearly independent.

By Theorem 3.7, the **general solution** is

$$\begin{aligned} y(t) &= c_1e^{\lambda t}\cos(\mu t) + c_2e^{\lambda t}\sin(\mu t), \\ &= e^{\lambda t}[c_1\cos(\mu t) + c_2\sin(\mu t)], \quad t \in \mathbb{R}. \end{aligned} \tag{3.17}$$

Example 3.16 Find the solution of the initial value problem

$$16y'' - 8y' + 145y = 0, \quad y(0) = -2, \quad y'(0) = 1\,.$$

Graph the solution and comment on its behaviour.

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3.3.3 Real, repeated roots

When $b^2 - 4ac = 0$ we have the repeated root

$$m_1 = m_2 = -\frac{b}{2a}\,.$$

Thus both roots yield the same solution

$$y_1(t) = e^{-\frac{bt}{2a}}$$

of (3.10). The general solution requires that we find a second linearly independent solution. We do this using a technique known as **reduction of order**.

3.3.4 Reduction of order[6]

[6] Reduction of order has been seen before in the MTH1002, part 2, lecture notes

The general idea is that, since $y_1(t)$ is a solution then so is $c\,y_1(t)$ for any constant c. However, this solution is not linearly independent.We generalise the method by replacing the constant c by a function $v(t)$ and we aim to determine $v(t)$ such that the product $v(t)y_1(t)$ is a linearly independent solution of (3.10).

We therefore postulate a solution

$$y = v(t)y_1(t) = v(t)e^{-\frac{bt}{2a}}$$

and substitute into (3.10), that is,

$$ay'' + by' + cy = 0$$

to determine $v(t)$. Using the product rule, we have

$$y' = v'e^{-\frac{bt}{2a}} - \frac{b}{2a}v\,e^{-\frac{bt}{2a}}$$

and

$$y'' = v''e^{-\frac{bt}{2a}} - \frac{b}{a}v'e^{-\frac{bt}{2a}} + \frac{b^2}{4a^2}v\,e^{-\frac{bt}{2a}}\,.$$

Now substitute into (3.10) to obtain

$$\left\{a\left[v'' - \frac{b}{a}v' + \frac{b^2}{4a^2}v\right] + b\left[v' - \frac{b}{2a}v\right] + cv\right\}e^{-\frac{bt}{2a}} = 0\,.$$

We cancel the (non-zero) exponential and rearrange the remaining terms to obtain

$$av'' + \left(\frac{b^2}{4a} - \frac{b^2}{2a} + c\right)v = 0\,.$$

However, we are considering the case of repeated roots, i.e. $b^2 = 4ac$. Hence the coefficient of v vanishes and we arrive at

$$v'' = 0\,.$$

Now set $w = v'$ to obtain

$$w' = 0\,,$$

i.e. we have reduced the order of the equation from second to first. The solution $w = c_1$ gives

$$v(t) = c_1 t + c_2$$

and so

$$y = c_1 te^{-\frac{bt}{2a}} + c_2 e^{-\frac{bt}{2a}}.$$

We have found that y is a linear combination of the two solutions

$$y_1(t) = e^{-\frac{bt}{2a}}, \quad y_2(t) = te^{-\frac{bt}{2a}}.$$

Their Wronskian is

$$\begin{aligned} W(t) &= \begin{vmatrix} e^{-\frac{bt}{2a}} & te^{-\frac{bt}{2a}} \\ -\frac{b}{2a}e^{-\frac{bt}{2a}} & \left(1 - \frac{bt}{2a}\right) e^{-\frac{bt}{2a}} \end{vmatrix} \\ &= e^{-\frac{bt}{a}}. \end{aligned}$$

Since $W[y_1, y_2](t)$ is never zero, the solutions y_1 and y_2 are a fundamental set of solutions and the **general solution** is

$$y = c_1 te^{-\frac{bt}{2a}} + c_2 e^{-\frac{bt}{2a}}, \quad t \in \mathbb{R},$$

i.e. one exponential solution corresponding to the repeated root and a second solution that is obtained by multiplying the exponential solution by t.

Example 3.17 Find the solution of the initial value problem

$$y'' - y' + \frac{y}{4} = 0, \quad y(0) = 2, \quad y'(0) = \frac{1}{3}.$$

Graph the solution and comment on its behaviour.

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Summary

We have shown that solutions of the second order, linear, homogeneous equation

$$ay'' + by' + cy = 0$$

are found by seeking solutions of the form $y = e^{mt}$. This leads to the characteristic polynomial

$$am^2 + bm + c = 0$$

and we must find its roots m_1 and m_2.

- If m_1 and m_2 are **real and not equal**, then the general solution is

$$\boxed{y = c_1 e^{m_1 t} + c_2 e^{m_2 t}}$$

- If m_1 and m_2 are **complex conjugates** $\lambda \pm i\mu$, then the general solution is

$$\boxed{y = c_1 e^{\lambda t} \cos(\mu t) + c_2 e^{\lambda t} \sin(\mu t)}$$

- If m_1 and m_2 are **real and equal**, then the general solution is

$$\boxed{y = c_1 e^{m_1 t} + c_2 t e^{m_1 t}}$$

3.4 Nonhomogeneous equations: Basic Theory

We now consider to the **nonhomogenous equation**

$$y'' + p(t)y' + q(t)y = g(t)\,, \tag{3.18}$$

where p, q and g are given continuous functions on the open interval $I : \alpha < t < \beta$. The equation

$$y'' + p(t)y' + q(t)y = 0\,, \tag{3.19}$$

with the same p and q but $g(t) \equiv 0$ is called the **corresponding homogeneous equation**.

Theorem 3.18 The general solution of the nonhomogeneous equation (3.18) can be written in the form

$$y(t) = c_1 y_1(t) + c_2 y_2(t) + y_p(t)\,, \tag{3.20}$$

where y_1 and y_2 are a fundamental set of solutions of the corresponding homogeneous equation (3.19), c_1 and c_2 are arbitrary constants, and y_p is any solution of the nonhomogeneous equation (3.18).

The method of solution for nonhomogeneous second order ODEs is then:

1. Find the general solution $c_1 y_1(t) + c_2 y_2(t)$ of the corresponding homogeneous equation. This is often called the **complementary solution** and is denoted by $y_c(t)$. We have already discussed how to find y_c when the homogeneous equation has constant coefficients.

2. Find a **particular solution** $y_p(t)$ of the nonhomogeneous equation. We will focus on how to do this below.

3. Add together y_c and y_p.

There are three methods for finding y_p that we discuss: the method of undetermined coefficients (§3.5), the method of variation of parameters (§3.6) and the method of reduction of order (§3.7).

3.5 Nonhomogeneous equations with constant coefficients[7]: Method of undetermined coefficients

[7] Nonhomogeneous ODEs with constant coefficients have been seen before in the MTH1002, part 2, lecture notes

Here we consider the nonhomogeneous equation with constant coefficients

$$ay'' + by' + cy = g(t)\,, \tag{3.21}$$

where a, b and c are constants and $a \neq 0$, and the corresponding homogeneous problem

$$ay'' + by' + cy = 0\,. \tag{3.22}$$

The method of undetermined coefficients requires that we make an 'intelligent guess' (a **trial solution**) about the form of the particular solution $y_p(t)$, but with the coefficients left unspecified. We then substitute the assumed form into the nonhomogeneous problem (3.21) and attempt to determine the coefficients such that y_p is a solution. If we cannot determine the coefficients then this means that our initial guess was incorrect and we must modify the assumption and try again. The method is used mainly when the homogeneous equation has constant coefficients (as in (3.22)) and the nonhomogeneous term is restricted to a small class of functions: polynomials, exponentials and sines and cosines.

3.5.1 Constructing the trial solution

- Find the general solution of the corresponding **homogeneous** problem, $y_c(t) = c_1 y_1(t) + c_2 y_2(t)$.
- Make sure that the nonhomogeneous term $g(t)$ involves nothing more than polynomials, exponentials and sines and cosines, or sums or products of such functions. If this is not the case, use the method of variation of parameters instead (see §3.6).
- Find a **particular solution of the nonhomogeneous problem** by assuming a trial solution of similar form to $g(t)$ (see the table below).
- The **general solution of the nonhomogeneous problem is** $y(t) = y_c(t) + y_p(t)$**.**

For IVPs: Use the initial conditions to determine the values of the arbitrary constants remaining in the general solution.

RHS $g(t)$	Trial solution $y_p(t)$
$A_k t^k + \cdots + A_1 t + A_0$	$(B_k t^k + \cdots + B_1 t + B_0)t^s$ where s is the number of times 0 is a root of the characteristic equation
$(A_k t^k + \cdots + A_1 t + A_0)e^{\alpha t}$	$(B_k t^k + \cdots + B_1 t + B_0)e^{\alpha t}t^s$ where s is the number of times α is a root of the characteristic equation
$(A_k t^k + \cdots + A_1 t + A_0)e^{\alpha t}\cos(\beta t)$ or $(A_k t^k + \cdots + A_1 t + A_0)e^{\alpha t}\sin(\beta t)$	$[(B_k t^k + \cdots + B_1 t + B_0)\cos(\beta t) + (C_k t^k + \cdots + C_1 t + C_0)\sin(\beta t)]e^{\alpha t}t^s$ where s is the number of times $\alpha + i\beta$ is a root of the characteristic equation

Example 3.19 Find the general solution of

$$y'' - 3y' - 4y = 3\exp(2t)\,.$$

Theorem 3.20 — Superposition principle for nonhomogeneous equations. A linear combination of nonhomogeneous terms $g_i(t)$, for $i = 1, \ldots, m$, implies a similar linear combination of particular integrals. That is, if $y_i(t)$, for $i = 1, \ldots, m$, are solutions of

$$y_i'' + p(t)y_i' + q(t)y_i = g_i(t)\,,$$

then

$$Y(t) = \sum_{i=1}^{m} c_i y_i(t)$$

is a solution of

$$Y'' + p(t)Y' + q(t)Y = G(t), \quad G(t) = \sum_{i=1}^{m} c_i g_i(t)\,,$$

for any set of constants c_i.

Activity 3.21 Prove Theorem 3.20.

3.6 Nonhomogeneous equations with variable coefficients: Variation of Parameters

Now we consider the nonhomogeneous problem with variable coefficients

$$y'' + p(t)y' + q(t)y = g(t)\,, \tag{3.24}$$

where p, q and g are given continuous functions on the open interval $I : \alpha < t < \beta$.

We begin by assuming[8] that we know the general solution

$$y_c(t) = c_1 y_1(t) + c_2 y_2(t) \tag{3.25}$$

of the corresponding homogeneous equation

$$y'' + p(t)y' + q(t)y = 0\,. \tag{3.26}$$

[8] Note that this is a major assumption because so far we have only considered solutions to the constant coefficient homogeneous equation. We meet a particular class of variable coefficient second order ODEs in §3.8.

Then, the idea is to replace the two constants c_1 and c_2 in (3.25) by two functions $u_1(t)$ and $u_2(t)$, say, so that we have

$$y = u_1(t)y_1(t) + u_2(t)y_2(t) \tag{3.27}$$

and we now aim to determine the unknown $u_1(t)$ and $u_2(t)$ in such a way that (3.27) is the solution of the nonhomogeneous problem (3.24).

We will do this by substituting (3.27) into (3.24), but since there are two unknowns and only one equation, we expect that there will be many possible choices of u_1 and u_2 that will suffice. Lagrange's method is to impose a (special[9]) second condition, giving two equations for two unknowns. We follow Lagrange's method.

[9] Lagrange showed how to choose this second condition in a special way to make the algebra much simpler than it otherwise would be.

First, we require the derivatives of y from (3.27) so that we can substitute them into (3.24). Thus, using the product rule, the first derivative is

$$y' = u_1(t)y_1'(t) + u_1'(t)y_1(t) + u_2(t)y_2'(t) + u_2'(t)y_2(t)\,.$$

Following Lagrange's method, we choose to make the terms involving u_1' and u_2' vanish, i.e. we set

$$u_1'(t)y_1(t) + u_2'(t)y_2(t) = 0 \tag{3.28}$$

and so the expression for y' becomes

$$y' = u_1(t)y_1'(t) + u_2(t)y_2'(t)\,. \tag{3.29}$$

Differentiating to find y'' we obtain

$$y'' = u_1(t)y_1''(t) + u_1'(t)y_1'(t) + u_2(t)y_2''(t) + u_2'(t)y_2'(t)\,. \tag{3.30}$$

Now we substitute y, y' and y'' into the nonhomogeneous equation (3.24). After gathering terms proportional to u_1 and u_2 we obtain

$$\begin{aligned} & u_1(t)\left[y_1''(t) + p(t)y_1'(t) + q(t)y_1(t)\right] \\ + \quad & u_2(t)\left[y_2''(t) + p(t)y_2'(t) + q(t)y_2(t)\right] \\ + \quad & u_1'(t)y_1'(t) + u_2'(t)y_2'(t) = g(t)\,. \end{aligned} \tag{3.31}$$

Now note that each of the terms in the square brackets is zero, because y_1 and y_2 satisfy the homogeneous problem (this is what we mean by (3.25)). Hence we are left with

$$u_1'(t)y_1'(t) + u_2'(t)y_2'(t) = g(t)\,. \tag{3.32}$$

Now we have two conditions, namely (3.28) and (3.32), that can be solved for the derivatives $u_1'(t)$ and $u_2'(t)$ of the unknown functions. We find

$$u_1'(t) = -\frac{y_2(t)g(t)}{W[y_1,y_2](t)}, \quad u_2'(t) = \frac{y_1(t)g(t)}{W[y_1,y_2](t)}\,,$$

where $W[y_1,y_2]$ is the Wronskian of y_1 and y_2 (note that since they are a fundamental set of solutions (see Theorem 3.7), their Wronskian is non-zero and we can divide by it). Finally we integrate to find

$$u_1(t) = -\int \frac{y_2(t)g(t)}{W[y_1,y_2](t)}\,dt + c_1, \quad u_2(t) = \int \frac{y_1(t)g(t)}{W[y_1,y_2](t)}dt + c_2\,.$$

Hence we have found the particular solution (3.27)

$$y_p(t) = -y_1(t)\int \frac{y_2(t)g(t)}{W[y_1,y_2](t)}\,dt + y_2(t)\int \frac{y_1(t)g(t)}{W[y_1,y_2](t)}dt$$

and, by Theorem 3.18, the general solution is

$$y = c_1y_1(t) + c_2y_2(t) + y_p(t)\,.$$

We summarise the result as a theorem.

Theorem 3.22 Consider the nonhomogeneous equation

$$y'' + p(t)y' + q(t)y = g(t)\,, \tag{3.33}$$

where p, q and g are given continuous functions on the open interval $I : \alpha < t < \beta$.

If the functions $y_1(t)$ and $y_2(t)$ are linearly independent solutions of the corresponding homogeneous equation

$$y'' + p(t)y' + q(t)y = 0\,, \tag{3.34}$$

then a particular solution of (3.33) is

$$y_p(t) = -y_1(t)\int_{t_0}^{t} \frac{y_2(s)g(s)}{W[y_1,y_2](s)}\,ds + y_2(t)\int_{t_0}^{t} \frac{y_1(s)g(s)}{W[y_1,y_2](s)}\,ds\,, \tag{3.35}$$

where $t_0 \in I$, and the general solution of (3.33) is

$$y = c_1y_1(t) + c_2y_2(t) + y_p(t)\,, \tag{3.36}$$

where c_1 and c_2 are constants.

Example 3.23 Use the method of variation of parameters to find the general solution of

$$y'' + 4y' + 4y = t^{-2}e^{-2t}, \quad t > 0\,.$$

3.7 Nonhomogeneous equations with variable coefficients: Reduction of Order

This is a powerful method to find the solution of a nonhomogeneous equation with variable coefficients given that one solution of the corresponding homogeneous problem is already known. We met a simpler version of it before in §3.3.4.

Suppose that we know one non-trivial solution $y_1(t)$ of the homogeneous equation

$$y'' + p(t)y' + q(t)y = 0 \tag{3.37}$$

and we want to find the solution of the nonhomogeneous equation

$$y'' + p(t)y' + q(t)y = g(t)\,. \tag{3.38}$$

We set

$$y(t) = v(t)y_1(t)$$

so that,

$$y' = vy_1' + v'y_1, \quad y'' = vy_1'' + 2v'y_1' + v''y_1\,.$$

Now substitute these derivatives into the nonhomogeneous equation (3.38) to obtain, after some rearranging

$$v[y_1'' + p(t)y_1' + q(t)y_1(t)] + v''y_1 + v'[2y_1' + p(t)y_1] = g(t)\,. \tag{3.39}$$

However, y_1 is a solution of the homogeneous equation (3.37) and so the first square bracket vanishes and we obtain

$$w'y_1 + w\,[2y_1' + p(t)y_1] = g(t)\,, \tag{3.40}$$

where we have set $w = v'$. This is now a first order equation for w which can be solved using an integrating factor.

Example 3.24 Solve

$$y'' - \frac{(4t+1)}{t}y' + \frac{2}{t}(2t+1)y = 4e^{2t}$$

on $t \in (0, \infty)$, given that one solution of the corresponding homogeneous equation is $y = e^{2t}$.

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3.8 Euler-Cauchy equations

These are a special class of homogeneous equation with specific variable coefficients. Euler-Cauchy equations take the form

$$t^2y'' + \alpha ty' + \beta y = 0, \quad \text{for } t > 0, \tag{3.41}$$

where α, β are constants.

We seek trial solutions of the form

$$y = t^r\,,$$

where r is a constant that is to be determined. Differentiating gives

$$y' = rt^{r-1}, \quad y'' = r(r-1)t^{r-2}\,.$$

Substituting into (3.41) yields

$$t^r\left[r(r-1) + \alpha r + \beta\right] = 0$$

and so we are required to solve the **auxiliary equation** (or **indicial equation**)

$$r(r-1) + \alpha r + \beta = 0\,. \tag{3.42}$$

The roots are

$$r_1, r_2 = -\frac{\alpha - 1}{2} \pm \frac{\sqrt{(\alpha-1)^2 - 4\beta}}{2}\,. \tag{3.43}$$

As we did for second order linear equations with constant coefficients, it is necessary to consider separately the cases in which the roots are real and different, real and equal, and complex conjugates.

3.8.1 Real and different roots

If $r_1 \neq r_2$ then we obtain the two solutions

$$y_1 = t^{r_1}, \quad y_2 = t^{r_2}.$$

Since

$$W[y_1, y_2] = (r_2 - r_1)t^{r_1+r_2-1},$$

which is non-zero for $t > 0$ and $r_1 \neq r_2$, it follows that the general solution is

$$y = c_1 t^{r_1} + c_2 t^{r_2}, \quad t > 0. \tag{3.44}$$

Note that if r is not a rational number, then t^r is defined by $t^r = e^{r \ln t}$.

Example 3.25 Solve

$$2t^2y'' + 3ty' - y = 0, \quad t > 0.$$

3.8.2 Real and equal roots

Here $r_1 = r_2 = -\frac{\alpha-1}{2} = r$, say, and we have only one solution

$$y_1 = t^r\,.$$

We then use the method of reduction of order (see §3.7) to determine the second solution. Writing

$$y = t^r v(t),$$

finding y' and y'' and substituting into the Euler-Cauchy equation (3.41) leads to (see activity 3.26)

$$t^2 v'' + t(2r+\alpha)v' = 0\,. \tag{3.45}$$

Equation (3.45) can be solved by separation of variables. We rewrite the equation as

$$\frac{v''}{v'} = -\frac{2r+\alpha}{t}\,,$$

since $t \neq 0$ and we assume that $v' \neq 0$. Integrating gives

$$\ln|v'| = -(2r+\alpha)\ln|t| + \ln c$$

where $\ln c$ is a constant. Hence

$$\begin{aligned} |v'| &= e^{-(2r+\alpha)\ln|t|+\ln c} = e^{\ln|t|^{-(2r+\alpha)}} e^{\ln c} \\ \Rightarrow v' &= d\,t^{-(2r+\alpha)}, \quad d = \pm c, \quad t > 0, \end{aligned} \tag{3.46}$$

using laws of logarithms and exponents.

Now, using the definition $r = r_1 = r_2$ we have

$$2r + \alpha = -2\frac{(\alpha-1)}{2} + \alpha = -1$$

and hence

$$v' = dt^{-1},$$

and integrating gives

$$v = d\ln t + f\,,$$

where f is a constant. Hence

$$y = t^r(d\ln t + f) = c_2 y_2 + c_1 y_1, \quad t > 0 \tag{3.47}$$

is the general solution, as it can be shown (see activity 3.27) that $W[y_1, y_2] = t^{2r-1}$ and y_1 and y_2 are linearly independent for $t > 0$.

Activity 3.26 Derive the ODE (3.45) for $v(t)$ above using the method of reduction of order.

Activity 3.27 Show that $y_1 = t^r$ and $y_2 = t^r \ln t$ are linearly independent for $t > 0$.

Example 3.28 Solve

$$t^2y'' + 5ty + 4y = 0, \quad t > 0\,.$$

3.8.3 Complex roots

Here the roots are complex conjugates, $r_1 = \lambda + i\mu$ and $r_2 = \lambda - i\mu$, say, with $\mu \neq 0$. First of all we need to define what we mean by t^r when r is complex. Recall that

$$t^r = e^{r \ln t}$$

when $t > 0$ and r is real. We use this same equation when r is complex. Thus

$$\begin{aligned} t^{\lambda+i\mu} &= e^{(\lambda+i\mu)\ln t} = e^{\lambda \ln t}\, e^{i\mu \ln t} = t^\lambda\, e^{i\mu \ln t} \\ &= t^\lambda \left[\cos(\mu \ln t) + i \sin(\mu \ln t)\right], \quad t > 0\,. \end{aligned} \tag{3.48}$$

The general solution is

$$y = c_1 t^{\lambda+i\mu} + c_2 t^{\lambda-i\mu}\,.$$

Note however that this is a complex-valued function. As was the case for the second order differential equation with constant coefficients with complex roots, it can be shown that the real and imaginary parts of $t^{\lambda+i\mu}$, namely

$$t^\lambda \cos(\mu \ln t) \quad \text{and} \quad t^\lambda \sin(\mu \ln t)\,,$$

are also solutions of the Euler-Cauchy equation (3.41) and their Wronskian

$$W[t^\lambda \cos(\mu \ln t), t^\lambda \sin(\mu \ln t)] = \mu t^{2\lambda-1}$$

is non-zero for $t > 0$ and $\mu \neq 0$. Hence the general solution for $t > 0$ can be written

$$y = c_1 t^\lambda \cos(\mu \ln t) + c_2 t^\lambda \sin(\mu \ln t), \quad t > 0\,. \tag{3.49}$$

Example 3.29 Solve

$$t^2y'' + ty' + y = 0, \quad t > 0.$$

Activity 3.30 As an alternative way of finding solutions to the Euler-Cauchy equation (3.41), use the change of variables $t = e^s$ and show that $y(t) = y(e^s) = Y(s)$ satisfies a linear ODE with constant coefficients. Hence find $y(t)$.

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3.9 Method of undetermined coefficients for nonhomogeneous Euler-Cauchy equations

A modified method of undetermined coefficients can be used to solve a nonhomogeneous Euler-Cauchy equation of the form

$$t^2y'' + \alpha ty' + \beta y = \gamma t^\delta$$

by considering trial solutions of the form $y = ct^\delta$ (for the non-resonant case). If that fails, try $y = ct^\delta \ln t$ or $y = ct^\delta (\ln t)^2$ (for the simple resonance and double resonance cases, respectively).